

Is a Decision Model a Reranker? A Four-Arm Benchmark on Real Documents

rerank-bench technical report — 2026-09-27

This report is the narrative arc of the whole benchmark: why it exists, how it was built, what happened at every stage from the first 10-query smoke to the final 175-query run, and what the combined results say. Companion documents carry the full depth: `evaluation-method.md` (protocol and formulas), `dataset-curation.md` (corpus and queries), `results/run1/report.md` (raw tables).

Abstract

Decision-style models (TypeSafe's SystemOne family, "Jev" here) are increasingly pitched as rerankers, but the claim is rarely measured against a dedicated cross-encoder. We built rerank-bench: 175 handmade queries over 1,992 chunks from 12 real documents (a novel, the US Constitution, a SCOTUS opinion, two OpenStax STEM textbooks, Dive into Deep Learning, six Wikipedia articles), with six query types including two-chunk reasoning, unanswerable negatives, document-furniture structural queries, and figure-content graphical queries. Four arms rerank the same top-30 nomic-embed-text candidates: the embedding order itself (baseline), NVIDIA's Llama Nemotron rerank VL 1B v2 (via OpenRouter, with page-image payloads on graphical queries), Jev in a one-call listwise shape, and Jev in a 30-call pointwise fan-out. On gold metrics (recall@1/5/10, MRR@10, nDCG@10 over 155 gold-bearing queries) reranking beats the embedding order decisively, Nemotron wins overall quality (recall@1 0.797 vs 0.532 baseline), Jev listwise is the best decision-model shape (0.632 at ~1s median and \$0.58/1k queries), and pointwise is dominated on every axis. Per type, all arms collapse on two-chunk reasoning queries (best recall@1 0.417) while page-image payloads pay off on graphical queries (0.880 vs 0.440 baseline). A position-swapped blind judge (Jev) with a Gemini Flash spot check (Cohen's kappa 0.17-0.37) corroborates the gold metrics where the two agree and is explicitly reported as too weak to overturn them where they disagree. We conclude: reranking is worth it, a dedicated cross-encoder still leads on quality, the listwise shape is the only sensible decision-model call pattern here, and multi-hop synthesis is the open problem no reranker in this study solves.

1. Introduction

Production retrieval stacks increasingly bolt an LLM in front of a vector search to reorder its candidates. Vendors of decision-style models — models whose native output is a structured choice with probabilities rather than free text — pitch them for exactly this job. Two claims deserve measurement:

1. Does LLM reranking beat the embedding order it sits on, by enough to justify its latency and cost?
2. Does a decision model (with a ranking-shaped prompt) match a purpose-built cross-encoder reranker?

Public benchmarks cover adjacent ground but not this comparison: BEIR/MS MARCO/ MTEB(r) score rerankers on short clean text with gold qrels and no LLM judge; BRIGHT brings reasoning-intensive long documents but stops at retrieval metrics; ViDoRe scores visual document retrieval but not reranking; the cross-encoder-vs-LLM-reranker literature compares on standard IR collections without decision models, real PDFs, or a judge protocol. rerank-bench fills that gap: decision model vs cross-encoder, on big real

documents including figures, scored by gold IR metrics and a swap-corrected blind judge with an inter-judge agreement statistic.

2. Benchmark design

The design (spec: `docs/specs/2026-09-26-rerank-bench-design.md`) fixes four arms behind one protocol — `rerank(query, candidates) -> (ranked_ids, scores, latency, cost, flags)` — so every arm sees identical inputs:

- **embed_only** — the cosine order of the top-30 candidates. The baseline; every other arm is measured as lift over it.
- **nemotron_vl** — NVIDIA's rerank cross-encoder via OpenRouter. On graphical queries each candidate also carries its PDF page rendered to PNG (150 dpi, with a DPI fallback ladder for oversized payloads).
- **jev_listwise** — one SystemOne `choice` call: all 30 candidates in the state (800-token cap each, truncation flagged), ranked by returned probabilities.
- **jev_pointwise** — 30 parallel SystemOne `nou1` calls ("does this passage answer the query?"), ranked by score. Tests the fan-out shape against the one-call shape.

Quality is gold IR metrics with strict empty-gold semantics (negative queries score None, never zero, and are excluded from aggregates by construction). The judge layer is blind (arm names never appear), position-swapped (disagreement between orders demotes to TIE and counts a flip), and self-checked (Gemini Flash re-judges 50 queries per pairing; Cohen's kappa reports how much the judge layer can be trusted — the designed counterweight to a Jev model judging Jev-produced rankings).

3. Dataset

175 queries, six types, over a 12-source corpus (licenses: 3 public domain, 2 CC-BY 4.0, 7 CC-BY-SA 4.0; provenance manifest committed):

type	n	tests	example (abbreviated)
verbatim	40	lexical match	which passage contains the line "Who-e debel you..."
paraphrase	40	semantic match	where does the text describe the whaler's weathered look...
reasoning	30	two-chunk synthesis	why does the narrator hesitate ... and what do the part-owners say
negative	20	discrimination	which amendment prohibits quartering soldiers... (topic absent)
structural	20	document furniture	on what page does the Transformer treatment begin?
graphical	25	figure/table content	which figure shows a whirlpool-in-a-tank image...

Negatives passed a two-pass verification (keyword co-occurrence scan, then domain-term scan with manual review of every hit); decoy hits were judged and kept as near-miss distractors where they make the query harder. Full procedure, balance tables, and limitations: `dataset-curation.md`.

4. Execution history (first run to last)

The benchmark was built test-first (75 tests) and executed in stages; each stage's finding shaped the next.

Stage 1 — wire verification. Both vendors' formats were verified live before any arm was coded (SystemOne `choice / noul` shapes; OpenRouter's rerank endpoint). This surfaced the first real finding: the pointwise wire format needs `instructions`, not `text`, on a `noul` question — a wrong guess here fails with an opaque 400.

Stage 2 — endpoint discovery. The base Nemotron slug returned 404 "No endpoints found" at every payload size on this account; the `:free` variant worked fully (verified to 62k-char payloads). The runner now probes both slugs before the first call. This finding saved the nemotron arm from silently recording garbage: the first smoke run had degraded all nemotron rows to candidate order before the probe was added.

Stage 3 — 10-query smoke. All four arms plus a first judge pass, on a 10-query slice: nemotron 0.900 recall@1 (vs embed 0.500), listwise 0.800, pointwise 0.600; graphical image payloads accepted with zero flags. The smoke also shook out three judge-layer bugs (chunk path, dataset key, partial-run handling) at 1/17th the cost of finding them in the full run.

Stage 4 — full run (run1). 175 queries x 4 arms in 45 minutes, 0 errors, \$0.26 total (budget: \$0.15-0.25 projected, \$1.00 abort). Judge pass: 33 minutes, all six pairings at n=175. Gemini spot check: 50 per pairing. One tooling crash (a missing default transport in the spot-check entry point) was fixed test-first and the stage rerun alone against persisted artifacts — the per-stage artifact design meant zero recomputation.

Stage 5 — whole-branch review. An independent reviewer recomputed every headline number from raw artifacts (exact match) and found four important defects — a dead DPI-ladder branch, error rows cached as successes, missing per-type reporting, no committed entry points for the judge/report stages — all fixed test-first before publication; none changed run1's numbers.

5. Results

5.1 Overall (155 gold-bearing queries per arm)

arm	recall@1	recall@5	recall@10	MRR@10	nDCG@10	p50 ms	p95 ms	\$/1k
embed_only	0.532	0.842	0.913	0.716	0.750	0	0	0
nemotron_vl	0.797	0.923	0.942	0.918	0.905	1935	18609	0
jev_listwise	0.632	0.923	0.948	0.817	0.837	982	1421	0.58
jev_pointwise	0.597	0.890	0.932	0.781	0.805	7687	9320	0.92

5.2 Per query type (recall@1 / nDCG@10)

type	embed_only	nemotron_vl	jev_listwise	jev_pointwise
verbatim (40)	0.725 / 0.837	0.950 / 0.950	0.750 / 0.876	0.750 / 0.876
paraphrase (40)	0.600 / 0.782	0.850 / 0.921	0.650 / 0.845	0.575 / 0.816
reasoning (30)	0.250 / 0.623	0.417 / 0.781	0.300 / 0.717	0.250 / 0.612
structural (20)	0.550 / 0.732	0.850 / 0.913	0.650 / 0.827	0.750 / 0.870
graphical (25)	0.440 / 0.724	0.880 / 0.948	0.800 / 0.915	0.680 / 0.851

5.3 Blind pairwise judging (Jev judge, both orders)

pairing	A wins	B wins	ties	flips	swap consistency
jev_listwise > jev_pointwise	27	5	143	81	0.54
nemotron_vl > jev_listwise	7	35	133	71	0.59
nemotron_vl > jev_pointwise	20	17	138	64	0.63
nemotron_vl > embed_only	55	7	113	66	0.62
jev_listwise > embed_only	69	3	103	51	0.71
jev_pointwise > embed_only	55	9	111	69	0.61

5.4 Gemini Flash spot check

pairing	kappa	n	flagged
jev_listwise > jev_pointwise	0.22	50	3
nemotron_vl > jev_listwise	0.17	50	4
nemotron_vl > jev_pointwise	0.23	50	2
nemotron_vl > embed_only	0.30	50	0
jev_listwise > embed_only	0.37	50	1
jev_pointwise > embed_only	0.20	50	4

6. What the combined results say

Finding 1 — reranking beats the embedding order; this is the benchmark's most solid result. Nemotron lifts recall@1 from 0.532 to 0.797 (+0.265), Jev listwise to 0.632 (+0.100). The blind judge independently agrees by landslides (55-7 and 69-3) — and those are exactly the pairings where the two judges agree best (kappa 0.30, 0.37). Two independent instruments, same direction, large margins. If a retrieval stack serves cosine order directly today, a reranker in front of it is worth its latency.

Finding 2 — the dedicated cross-encoder still leads on quality. Nemotron's margin over the best decision-model arm is +0.165 recall@1, +0.10 MRR@10, and it leads every per-type cell. The price is the tail: p95 18.6s (free-endpoint queueing) against listwise's 1.4s. Quality-critical offline work should take the cross-encoder; latency-sensitive serving has a real trade to make.

Finding 3 — between the two decision-model shapes, listwise dominates. One structured call beats 30 fan-out calls on quality (0.632 vs 0.597 recall@1), median latency (982 vs 7687 ms), and cost (\$0.58 vs \$0.92 per 1k). Pointwise is not slightly worse; it is worse on every axis while making 30x the calls. The decision-model reranker story, where it works at all, is a listwise story.

Finding 4 — the judge layer is weak evidence, and the benchmark says so. Kappa between the Jev judge and its Gemini spot check is 0.17-0.37 — slight to fair agreement. Where judge and gold agree (both vs embed_only) the judge corroborates. Where they contradict — the judge prefers jev_listwise over nemotron 35-7 while gold says the opposite by a wide margin — the kappa says do not trust the judge, and the benchmark reports it that way rather than headlining it. Ties dominate (58-82%) because all arms rerank the same 30 candidates; this is structural, and the swap-consistency column (0.54-0.71) plus the spot check quantify how noisy the instrument is. The design lesson generalizes: an LLM-as-judge layer without an agreement statistic is unfalsifiable.

Finding 5 — reasoning queries are the open problem; graphical queries are where multimodality pays. Two-chunk synthesis collapses every arm (best recall@1 0.417, nDCG@10 0.781): reranking isolated candidates cannot assemble an answer that spans two passages — a retrieval-then-synthesize architecture problem, not a reranker-quality problem. Conversely the graphical slice shows the page-image payload earning its bytes: Nemotron with images reaches 0.880 recall@1 against 0.440 for the text-only baseline — and the text-only Jev listwise arm's 0.800 shows the figure captions were retrievable but badly ordered, i.e. the baseline's graphical failure is a ranking failure that any strong reranker largely fixes, with the image payload adding the last increment.

7. Threats to validity

- **Single run, no variance estimates.** With n=155 the headline gaps (0.10-0.27) are far beyond noise; ~0.05 differences (per-type cells, n=20-40) are within it.
- **Self-preference in the judge.** A Jev model judged Jev-produced rankings; the Gemini spot check and kappa exist precisely to price this, and Finding 4 prices it: low.
- **One author's gold labels.** No inter-annotator agreement; gold errors would shift absolute numbers but hit all arms identically (shared candidates).
- **Endpoint specifics.** Nemotron ran on the :free variant (the base slug has no endpoint on this account); its latency includes free-tier queueing. Jev pricing (\$0.042/Mtok input) is TypeSafe's current rate.
- **Corpus blind spots.** Mangled PDF math extraction, English-only, no code or handwritten sources.

8. Conclusion

Measured on real documents with handmade queries and a self-checking judge layer: (1) rerank your candidates — the lift over embedding order is large and doubly confirmed; (2) a purpose-built cross-encoder reranker still beats a decision model prompted into reranking, by a wide margin, at the cost of a long latency tail; (3) if you do use a decision model, use one listwise call — the pointwise fan-out shape is dominated everywhere; (4) no reranker in this study solves two-chunk reasoning, which points at synthesis rather than ranking as the next bottleneck; and (5) page-image payloads measurably pay for themselves on figure-grounded queries. The full artifact chain — corpus manifest, dataset literals, run rows, judge verdicts, spot checks, reports — is committed, and every stage rebuilds from the previous one.

References

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